

Pathfinder

From seminar questions
to research

Vincent Danos

CNRS, ENS

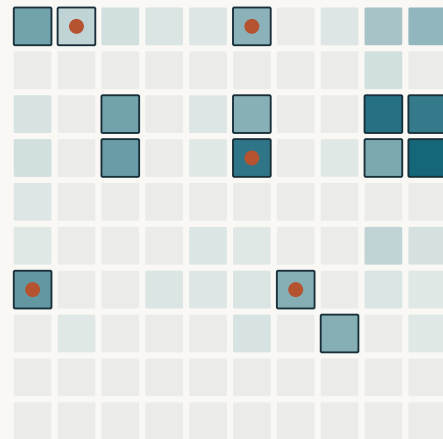
Elham Kashefi

CNRS, LIP6
University of Edinburgh

William Waites

University of Edinburgh

github.com/vd1/pathfinder



Ten papers by ten papers.
Every cell is a possible encounter.

2026-09-17

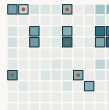
IQ lab, LIP6 17 September 2026

Pathfinder

From seminar questions
to research

Vincent Danos **Elham Kashefi** **William Waites**
CNRS, ENS CNRS, LIP6 University of Edinburgh
University of Edinburgh

github.com/vd1/pathfinder



Ten papers by ten papers.
Every cell is a possible encounter.

Timing: 15 seconds. Introduce the project and the authors. This is a programme for generating and investigating connections between research papers, with a small pilot and larger quantum-facing experiments. The main talk is approximately fifteen minutes, leaving five minutes for discussion.

Convention: mini-abstracts

Q mechanism design

Q4 Incentives to report what one knows

Agents may profit by hiding what they know.
Rewards can make truthful reporting worthwhile.

Q7 Sharing resources

Participants compete for limited resources.
Messages and payments aim to share them well.

Q3 AI agents in a market

AI agents buy and sell in a simulated market.
Their performance is compared with humans.

P systems of cooperating agents

P6 An outside judge for a team

A team works together; credit is hard to assign.
An outside AI judges contributions from images.

P1 Learning about bad outcomes

Learn to act while controlling bad outcomes.
Estimate losses in the worst fraction of cases.

P3 A critic for the code reviewer

An AI reviews code; a critic challenges its review.
The coding agent then revises the program.

2026-09-17

Convention: mini-abstracts

Timing: 45 seconds. Explain the convention before the first scientific example. Each teal box contains the paper's corpus label and a miniature abstract. These are reminders of the source papers, not statements of Pathfinder's contribution. They recur every time we discuss those papers, so the audience does not need to memorise Q4 or P6. Between the papers is a seed question, not an established claim. The scientific story is how the investigation moves from that question to a result, an obstruction, or a concrete missing experiment.

Convention: mini-abstracts	
Q mechanism design	P systems of cooperating agents
Q4 Incentives to report what one knows Agents may profit by hiding what they know. Rewards can make truthful reporting worthwhile.	P6 An outside judge for a team A team works together; credit is hard to assign. An outside AI judges contributions from images.
Q7 Sharing resources Participants compete for limited resources. Messages and payments aim to share them well.	P1 Learning about bad outcomes Learn to act while controlling bad outcomes. Estimate losses in the worst fraction of cases.
Q3 AI agents in a market AI agents buy and sell in a simulated market. Their performance is compared with humans.	P3 A critic for the code reviewer An AI reviews code; a critic challenges its review. The coding agent then revises the program.

Making connexions ...



P6 An outside judge for a team
A team works together; credit is hard to assign.
An outside AI judges contributions from images.

Q4 Incentives to report what one knows
Agents may profit by hiding what they know.
Rewards can make truthful reporting worthwhile.

2026-09-17

Making connexions ...

Timing: 45 seconds. This is a made-up seminar exchange, not the history of how these papers met. P6 uses an external AI judge; it does not assume strategic participants report truthfully. Q4 supplies conditional incentives for truthful reporting. The natural question is whether the latter can replace the former. Pathfinder tries to systematise both the generation of such questions and the research that follows them. We will return to what this particular connection actually established.



What if we asked across an entire collection?

Mechanism design × agent cooperation

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
Q1	1925	600*	300	150	75	1500*	0	75	1050	1400
Q2	0	0	0	0	0	0	0	0	300	0
Q3	200	0	1925	0	75	1575	0	0	3300	3000
Q4	300	0	2100	0	25	3150*	0	25	1800	3575
Q5	75	0	0	9	0	0	0	0	0	0
Q6	25	0	0	0	100	50	0	0	600	240
Q7	2200*	0	0	120	75	150	1650*	0	150	25
Q8	0	25	0	0	0	200	0	1650	0	50
Q9	0	0	0	0	0	0	0	0	0	0
Q10	0	0	0	0	0	0	0	0	0	0



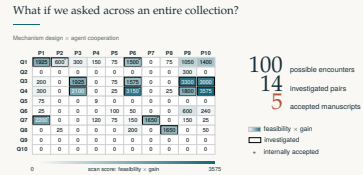
100 possible encounters
14 investigated pairs
5 accepted manuscripts

- feasibility × gain
- investigated
- internally accepted

2026-09-17

What if we asked across an entire collection?

Timing: 45 seconds. The heatmap is the root 10 by 10 experiment. The scan judges possible research connections, not textual similarity. Thirteen pairs met the score threshold of 1500; an already-started pair scoring 600 was retained, giving fourteen investigations. Five manuscripts were internally accepted. Point out Q4P6, at row four and column six. The Q1P7 zero is the separate 16 September rescan; the original response was unparseable. Selection and acceptance are not an estimate of a discovery rate, and we have not assessed what alternative selection policies would produce.



Scan. Research. Edit.



The output may be a result, an obstruction, a mistake in either paper, or an account of what is missing to pursue the connexion.

2026-09-17

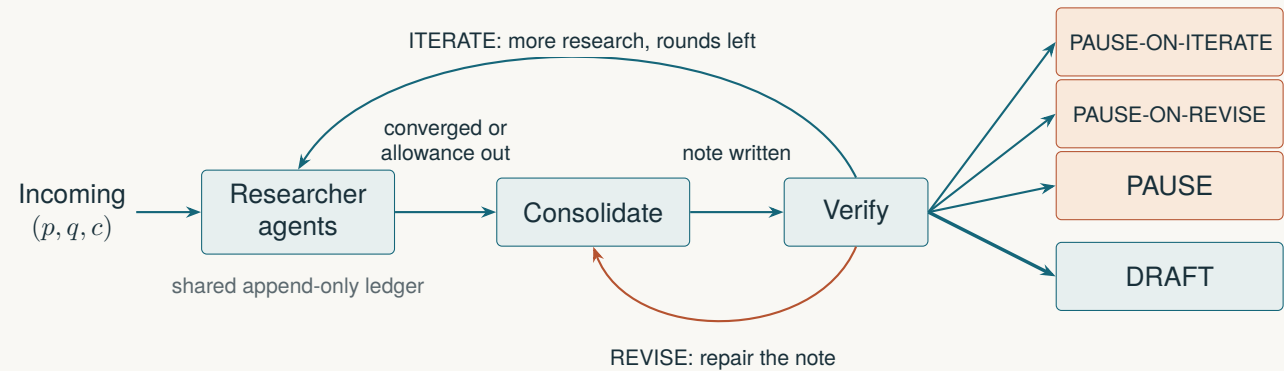
Scan. Research. Edit.



The output may be a result, an obstruction, a mistake in either paper, or an account of what is missing to pursue the connexion.

Timing: 45 seconds. Explain only the phases, not the formal state notation. Scan generates the seminar question. Research tests it and can reformulate it. Both DRAFT and PAUSE outcomes enter editing. The diagram shows their main destinations: a publication manuscript for DRAFT, a readable human-centric account for PAUSE, including capped pauses. The implementation also permits a readable account after DRAFT; the diagram emphasises the manuscript route for that outcome. Editing is not another research pass. Successful manuscript review and a successful build are both required for acceptance. The agent-facing specification and conformance tests are in the paper.

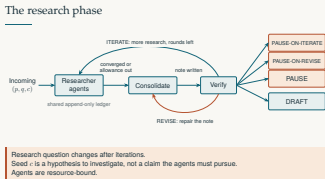
The research phase



Research question changes after iterations.
Seed c is a hypothesis to investigate, not a claim the agents must pursue.
Agents are resource-bound.

2026-09-17

The research phase



Timing: 1 minute. The researchers share an append-only ledger and can disagree. Their readiness is invalidated by subsequent substantive work. The consolidation is reviewed by a fresh verifier. Iterate means more research; revise means repair the note. The revise arc is below the note-written arc. A pause-on-iterate records an exhausted iteration allowance, not necessarily a dead end. We have discussed adding a separate scientific assessment, PAUSE-PLUS versus PAUSE-MINUS; that is a proposed refinement, not an implemented state transition in this diagram.

Practicalities

Nothing is lost

Every step is saved as it happens.

After a crash, work resumes where it stopped.

What an agent reads, in order

The two papers	The shared ledger, which only grows	This call's instructions
----------------	-------------------------------------	--------------------------

the same as last time, so cheap to reread

The machinery keeps the work safe and the bill predictable. It makes no scientific judgement.

Practicalities

Timing: 45 seconds. These are operational mechanisms, not additional scientific verdicts. The runner keeps artifacts, a shared ledger and stage state. Recovery resumes from a recorded boundary; it does not promise to recover an interrupted model’s internal state or avoid every duplicate call. Transport failure and unreadable output are handled separately from a scientific pause, with bounded retries and explicit intervention where needed. Receipts track call-level tokens and time; the older pilot lacked complete cache counters, as the economics slide explains. Budget admission uses available accounting and estimates, and in-flight work can exceed that estimate. For calls using inline context, common paper text comes first, then the growing ledger, then call-specific instructions. Prefix reuse depends on provider cache behaviour; it is not guaranteed. Researchers also read new ledger entries before changing direction or declaring ready.

Practicalities

Nothing is lost

Every step is saved as it happens.
After a crash, work resumes where it stopped.

Spending is bounded

Every call records its time and its cost.
Each loop has a limit, and the budget is checked before new work starts.

Reading is reused

Every call starts with the same material.
What the model has already read costs much less to read again.

What an agent reads, in order

The two papers

The shared ledger, which only grows

This call's instructions

the same as last time, so cheap to reread

The machinery keeps the work safe and the bill predictable. It makes no scientific judgement.

Q4P6: when can we dispense with the judge?

Q4 Incentives to report what one knows

Agents may profit by hiding what they know.
Rewards can make truthful reporting worthwhile.

P6 An outside judge for a team

A team works together; credit is hard to assign.
An outside AI judges contributions from images.

Seed

Replace the judge with the participants' own reports.

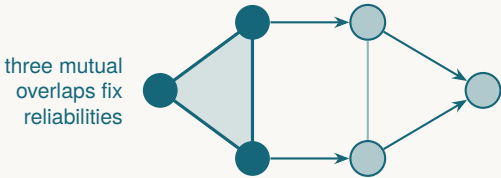
Obstacle

Truthful reports can still be insufficient for ranking.

Conditions

Informative observations. Structured errors.
Enough shared comparisons.

Conditional result



Under these conditions, peer reports identify the model's ranking without an external judge.

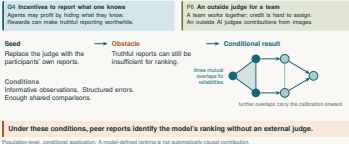
Population-level, conditional application. A model-defined ranking is not automatically causal contribution.

2026-09-17

Q4P6: when can we dispense with the judge?

Timing: 1 minute 45 seconds. Q4 addresses incentives, not whether the reports contain enough information to rank contributions. Under stable symmetric errors, observers that are better than chance, and conditional independence on a shared comparison, overlaps let us identify their reliabilities. A calibration triangle anchors this, and connectivity propagates it. A connected comparison graph then identifies the Bradley–Terry ranking up to an additive constant. This uses established single-coin calibration ideas, not a new general calibration theorem. The truthful-report mechanism still needs its own belief assumptions. The result is a conditional no-judge variation on P6, not a correction to a paper that already relied on strategic user reports.

Q4P6: when can we dispense with the judge?



Q7P1: a tale of two runs

Q7 Sharing resources

Participants compete for limited resources.
Messages and payments aim to share them well.

P1 Learning about bad outcomes

Learn to act while controlling bad outcomes.
Estimate losses in the worst fraction of cases.

Shared seed: adapt resource sharing to participants who care about bad outcomes.

First run



Rerun

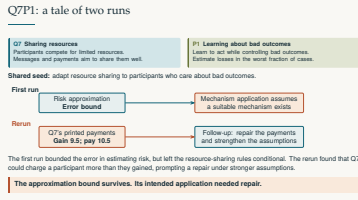


The first run bounded the error in estimating risk, but left the resource-sharing rules conditional. The rerun found that Q7 could charge a participant more than they gained, prompting a repair under stronger assumptions.

The approximation bound survives. Its intended application needed repair.

2026-09-17

Q7P1: a tale of two runs



Timing: 1 minute 45 seconds. The first paper establishes error control for an expected empirical, smoothed risk approximation, and near-optimal true-risk choices over the same feasible set. Its mechanism application is conditional. The independent rerun finds incompatible conditions and a failure of the printed payment rule in Q7, not P1. In an equilibrium example, a participant gets benefit 9.5 and pays 10.5. This does not refute a conditional theorem whose premises fail; it shows that the printed construction cannot support the desired application. A subsequent paper follow-up adds an opponent-only payment term and stronger curvature and feasibility assumptions. It proves guarantees for a fixed surrogate, with error control relative to true risk. The old approximation bound survives because it did not depend on those payments. Efficient learning in practice remains open. These are two runs, not a single uninterrupted chain of reasoning.

Q3P3: a promising question, an unfinished test

Q3 AI agents in a market

AI agents buy and sell in a simulated market. Their performance is compared with humans.

P3 A critic for the code reviewer

An AI reviews code; a critic challenges its review. The coding agent then revises the program.

Seed: spot a rush to finish

Q3 finds urgency in traders' language just before a trade. P3's reviewers sometimes agree too soon.

Could Q3's language analysis warn when P3's reviewers are rushing to agree?

Shift: test simpler agents

The researchers found that simple random traders could do well under Q3's rules.

This suggested a different question for P3: **does the critic help, or do extra rounds of review explain the gain?**

Test: compare fairly

Compare a reviewer working alone with a reviewer helped by a critic. Let both review and repair repeatedly, on the same tasks, with equal computing budgets.

The required materials were unavailable.

Q3 inspired the test; it did not establish the answer for P3. Neither the seed nor the revised connexion was demonstrated.

Original outcome: PAUSE-ON-ITERATE. Missing evidence, not proof that the critic is ineffective.

2026-09-17

Q3P3: a promising question, an unfinished test

Timing: 1 minute 30 seconds. The original seed transfers Q3's analysis of traders' language to P3's premature reviewer—critic agreement. Urgency before a trade is not itself evidence of premature agreement in code review; that proposed connection was not established. The researchers then changed direction. Their constrained random-trader benchmark performed well under Q3's stated market rules, conditional on the LLM traders complying with the same constraints. This motivated separating the value of agents' information from the rules governing their interaction. On the P3 side, some baselines revise once, while adversarial review and another stronger method revisit revised work. The proposed control compares repeated single review and adversarial review with equal resources, plus a random critic matched to disagreement rates. The supplied record lacked the tasks, execution setup and access needed to reproduce and extend the reported experiments. A new benchmark would test a related hypothesis, not explain those results. The first thread exhausted its iteration cap. A further continuation was requested, but no completed new outcome was available when these slides were built.

Q3P3: a promising question, an unfinished test

Q3 AI agents in a market
AI agents buy and sell in a simulated market. Their performance is compared with humans.

P3 A critic for the code reviewer
An AI reviews code; a critic challenges its review. The coding agent then revises the program.

Seed: spot a rush to finish
Q3 finds urgency in traders' language just before a trade. P3's reviewers sometimes agree too soon.
Could Q3's language analysis warn when P3's reviewers are rushing to agree?

Shift: test simpler agents
The researchers found that simple random traders could do well under Q3's rules.
This suggested a different question for P3: **does the critic help, or do extra rounds of review explain the gain?**

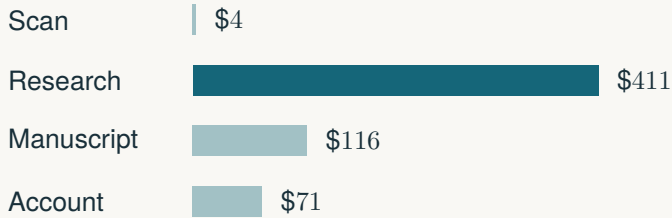
Test: compare fairly
Compare a reviewer working alone with a reviewer helped by a critic. Let both review and repair repeatedly, on the same tasks, with equal computing budgets. The required materials were unavailable.
Q3 inspired the test; it did not establish the answer for P3. Neither the seed nor the revised connexion was demonstrated.

Original outcome: PAUSE-ON-ITERATE. Missing evidence, not proof that the critic is ineffective.

Breadth is cheap. Investigation dominates.

Economics of the 10×10 pilot

\$601 recorded-token scenario



51.0M recorded input tokens; 1.82M output.
Flat GPT-6 Standard rates: \$10/M input, \$50/M output.

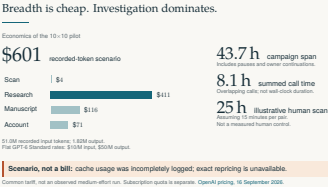
Scenario, not a bill: cache usage was incompletely logged; exact repricing is unavailable.

Common tariff, not an observed medium-effort run. Subscription quota is separate. [OpenAI pricing, 16 September 2026.](#)

2026-09-17

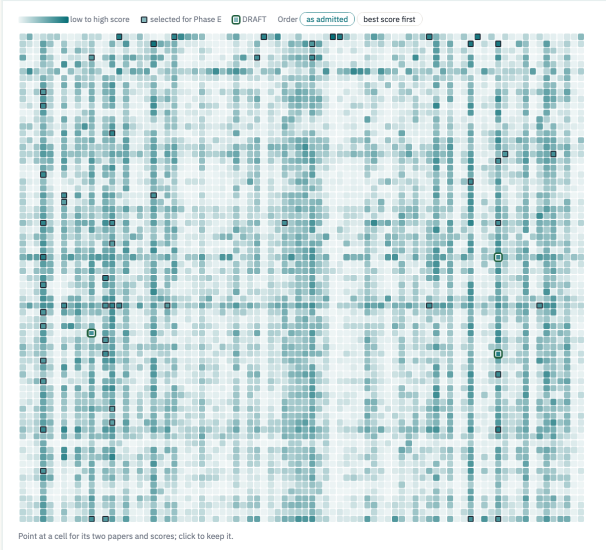
Breadth is cheap. Investigation dominates.

Timing: 1 minute. All displayed model dollar figures use the same GPT-6 Standard short-context tariff on the input and output tokens actually recorded. This deliberately simple scenario treats recorded input as uncached and omits unrecorded cache usage. Historical Claude cache reads and writes were not retained; Codex cached shares are also missing. Long-context adjustments and separately billed tools cannot be reconstructed. Thus 601 dollars is neither a measured GPT-6 bill nor a reliable upper or lower bound on one. It is a transparent illustration of pricing the available token counters. The 330 calls span 43.7 hours but sum to 8.1 agent-hours. Neither number establishes a human speedup. The human scan estimate assumes 15 minutes for each of 100 pairs, and is not matched for quality. Choosing medium effort does not predict token use.



Scale-up: the Julien Michel validation experiment

Quantum methods for computational chemistry



71 × 82
5,822 scored pairs

59 pairs selected for research
98 finished threads (51 PAUSE, 44 CONTINUE)
3 DRAFT outcomes

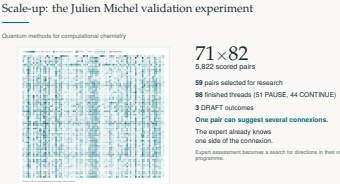
One pair can suggest several connexions.

The expert already knows
one side of the connexion.

Expert assessment becomes a search for directions in their own
programme.

2026-09-17

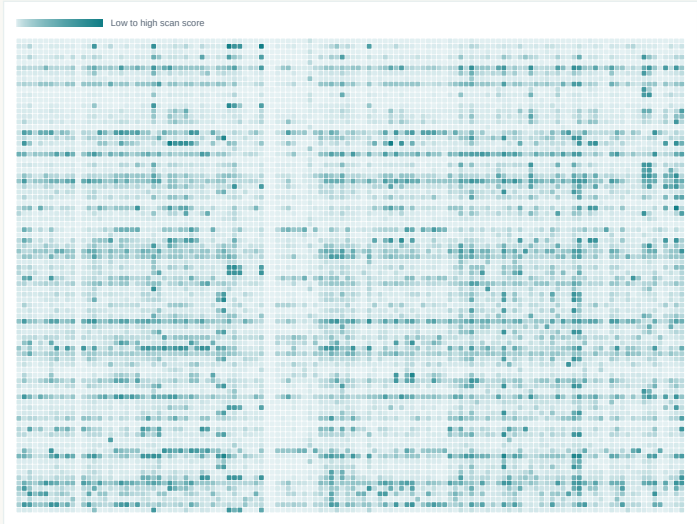
Scale-up: the Julien Michel validation experiment



Timing: 1 minute. The admitted release has 82 papers on each side; the original working inventories contain additional excluded rows. The supplied dashboard screenshot replaces the earlier reconstructed heatmap. Its legend marks selection for Phase E and DRAFT outcomes. Do not attach the old single-instrument coverage figures or snapshot date to this screenshot. Selection and DRAFT are workflow outcomes, not independent scientific validation. This is a different corpus and protocol from the 10 by 10 pilot. The expert's familiarity with one side makes evaluating candidate connections less costly and more directly relevant. The 59 selected pairs and 98 finished E-prime threads count different units. The thread outcomes are 51 paused, 44 continue and 3 draft-ready. CONTINUE ends a run, not the research question. A pair can suggest several connexions, and an investigation can also have follow-up rounds. The 98-thread count is not a count of distinct connexions or terminal research outcomes. Human validation asks whether a note changes what an expert understands or would investigate next: is the connection already familiar, are its assumptions meaningful, and does its conclusion help? This is an evaluation question, not a claim that the reported notes have passed that evaluation. Expert judgment complements proofs and experiments; it does not replace them. A supported obstruction can be useful even when the research ends at PAUSE rather than DRAFT.

Scale-up in quantum: QSL meets vendor applications

124 vendor-side paper artefacts (columns)



10,912 candidate pairs

100 pairs in the original selection

Fully scored by two judges (Opus 5 and gpt-5.6).

Global rank correlation: **0.740**.



Pathfinder: hypothesis generation across the quantum software-hardware boundary
Danos, Kashefi, Waites
Companion manuscript, 2026

Later E-prime campaign

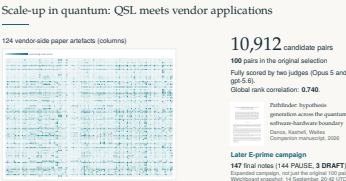
147 final notes (144 PAUSE, 3 DRAFT)

Expanded campaign, not just the original 100 pairs.
Watchboard snapshot: 14 September, 20:42 UTC.

2026-09-17

Scale-up in quantum: QSL meets vendor applications

Timing: 1 minute. Use the companion manuscript’s frozen 88 by 124 universe, not the approximate 150 by 150 live-corpus ambition. Both Opus 5 and GPT-5.6 Sol scored 10912 pairs. The heatmap displays Opus scores; its scale differs from the pilot’s feasibility and gain axes. The paper reports global Spearman correlation 0.740 but only 12–36 overlap between exact top-100 lists when cutoff ties are resolved. Thus broad rank agreement does not mean agreement about the best candidates. The paper audits selected candidate experiments; it does not report their execution or a measured discovery rate. Source: the local companion manuscript and its frozen-corpus ledger.



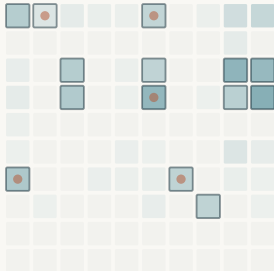
Questions

How much of creativity is connexion?
What is the density $d(Q, P)$ of $Q \times P$ in good ideas?
Can agents also help choose which corpora to bring together?
How to trade-off search space restriction and loss rate?

If agents can make the connexions and also investigate what follows ...

What are the other things humans do in science?

Choosing questions. Inventing concepts.
Judging significance. Applying for funding.



2026-09-17

Questions

How much of creativity is connexion?
What is the density $d(Q, P)$ of $Q \times P$ in good ideas?
Can agents also help choose which corpora to bring together?
How to trade-off search space restriction and loss rate?

If agents can make the connexions and also investigate what follows ...

What are the other things humans do in science?

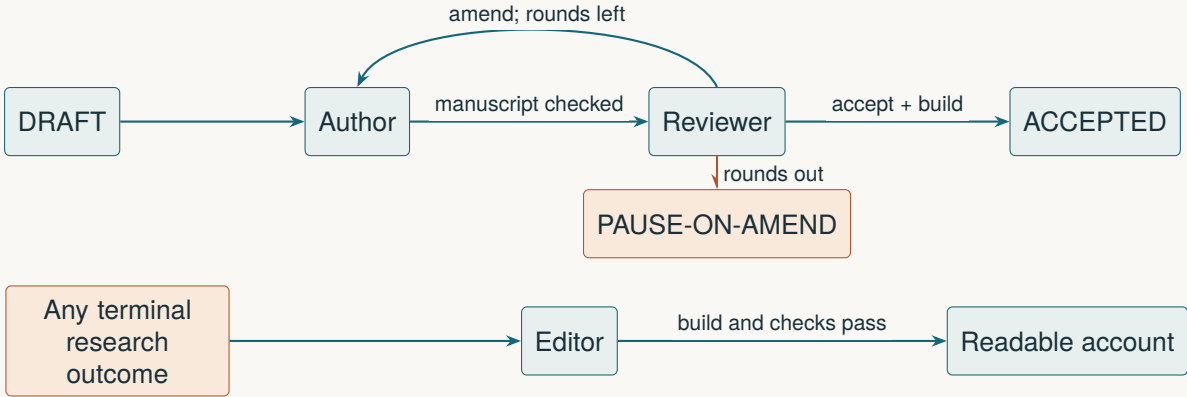
Choosing questions. Inventing concepts.
Judging significance. Applying for funding.



Timing: 1 minute, then five minutes of discussion. Pathfinder explores creativity through connections; it does not establish that creativity reduces to connections. Finding a candidate, establishing its consequences, and judging those consequences important are different achievements. If some stages become cheap, the scarcity may move to selecting problems, introducing concepts, or developing shared mathematical taste. Ask also whether agents can select useful neighbouring corpora, rather than only working within a rectangle chosen by a human. The mathematicians question is intentionally provocative, not a prediction of a date or a claim demonstrated by this pilot. Close with: If agents can make and investigate the connections, who decides which mathematics is worth doing?

Backup: editing preserves the research outcome

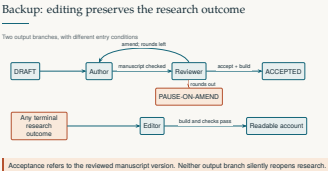
Two output branches, with different entry conditions



Acceptance refers to the reviewed manuscript version. Neither output branch silently reopens research.

2026-09-17

Backup: editing preserves the research outcome



Backup. Build and reference failures can trigger bounded repairs or an operational blockage. Those branches are omitted from this diagram for readability. The full note specifies them and provides conformance tests. Increasing an allowance does not itself reopen a terminal investigation; an owner continuation should preserve the previous outcome and record its new scope.

Backup: what Q4P6 actually assumes

Q4 Incentives to report what one knows

Agents may profit by hiding what they know.
Rewards can make truthful reporting worthwhile.

- Truthful reporting under Q4’s own incentive and belief assumptions.
- Observers are better than chance, with stable symmetric errors.
- Their errors are conditionally independent on the same latent comparison.
- Overlapping observations contain a calibration triangle and connect onward.
- The agent-comparison graph is connected; the ranking model is specified.

P6 An outside judge for a team

A team works together; credit is hard to assign.
An outside AI judges contributions from images.

A population-identification result, not a finite-sample guarantee or a theorem that these conditions arise automatically.

Backup: what Q4P6 actually assumes

Backup. Pairwise observer moments factor as products of reliabilities. A triangle fixes positive reliabilities, connected overlaps propagate them, and corrected means identify pairwise comparison probabilities. The Bradley–Terry model then yields contribution scores up to an additive constant. These are established calibration ingredients used in a conditional interface, not a new universal identification method. Private peer beliefs, calibration bootstrapping and causal meanings of contribution remain separate issues.

Q4 Incentives to report what one knows

Agents may profit by hiding what they know.
Rewards can make truthful reporting worthwhile.

P6 An outside judge for a team

A team works together; credit is hard to assign.
An outside AI judges contributions from images.

- Truthful reporting under Q4’s own incentive and belief assumptions.
- Observers are better than chance, with stable symmetric errors.
- Their errors are conditionally independent on the same latent comparison.
- Overlapping observations contain a calibration triangle and connect onward.
- The agent-comparison graph is connected; the ranking model is specified.

A population-identification result, not a finite-sample guarantee or a theorem that these conditions arise automatically.

Backup: what the dollar scenario does and does not say

Original 330-call cohort; later runs excluded

Stage	Calls	Summed call time	Scenario USD
Scan	106	18.3 min	3.75
Research + consolidation + verification	168	6.07 h	410.56
Manuscript preparation + review	38	1.23 h	116.11
Readable accounts	17	28.7 min	70.52
Probe	1	4.4 s	0.13

GPT-6 Standard, per million tokens: input \$10; cache read \$1; cache write \$12.50; output \$50.

Only input and output counters were retained in these receipts. Missing cache and per-request context data prevent exact normalisation.

Scenario applies \$10/M and \$50/M to recorded counters only. Above 272K input tokens, the full request has higher rates. Medium effort is not a multiplier that reconstructs a counterfactual run.

2026-09-17

Backup: what the dollar scenario does and does not say

Backup. The displayed stage values are an arithmetic scenario, not actual model charges. Claude input counters exclude unrecorded cache activity; Codex input may include cached tokens. Consequently the missing terms do not have a known net direction. A correct future cost logger needs disjoint uncached input, cache reads, cache writes, billed output including reasoning, request-level context length, and separately billed tools. Source: OpenAI model pricing checked 16 September 2026, and the original root receipts frozen into the slide assets.

Backup: what the dollar scenario does and does not say

Original 330-call cohort; later runs excluded

Stage	Calls	Summed call time	Scenario USD
Scan	106	18.3 min	3.75
Research + consolidation + verification	168	6.07 h	410.56
Manuscript preparation + review	38	1.23 h	116.11
Readable accounts	17	28.7 min	70.52
Probe	1	4.4 s	0.13

GPT-6 Standard, per million tokens: input \$10; cache read \$1; cache write \$12.50; output \$50.

Only input and output counters were retained in these receipts. Missing cache and per-request context data prevent exact normalisation.

Scenario applies \$10/M and \$50/M to recorded counters only. Above 272K input tokens, the full request has higher rates. Medium effort is not a multiplier that reconstructs a counterfactual run.

Backup: why did we pause?

Proposed review distinction, separate from the stopping mechanism

PAUSE-PLUS

An actionable continuation exists.

Name the missing material or experiment, the next step, and what its outcomes would establish.

PAUSE-MINUS

Evidence undermines this direction.

Explain what was ruled out and why, without claiming that every possible connection is exhausted.

An exhausted iteration budget is a stopping reason, not a scientific assessment. Allow “undetermined” too.

2026-09-17

Backup: why did we pause?

Backup. This is a proposed addition to the final review, not a claim that the implementation already has these states. The two dimensions should remain separate: where the runner stopped, and what the researchers learned about the prospects of continuing. PAUSE-PLUS must name an actionable next step; vague optimism is insufficient. Budget exhaustion may leave the assessment undetermined.

